



PLATFORM ENGINEERS • RELEASE OPERATORS

DEVOPS MANUAL

CI, Docker, Nuitka, metadata crypto, GCP, security

Open-source Pine Script evaluation framework



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OVERVIEW

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title: "DevOps" description: "CI matrices, containers, Nuitka LSP binaries, metadata crypto, Cloud Run, and operational invariants for PYNE."

DEVOPS

Abstract

PYNE is a **multi-surface product**: pure-Python library + Click CLI (`pyne` / alias `pynescrpt`), pygls Language Server (`pyne-lsp`), Flask Pro API, VS Code extension (`hoox-sh.pyne` 0.4.4; the Python package is 0.6.7), and Docker targets (`api` / `cli` / `lsp`). Edge workers and the AXIS charting PWA are **sister repos** — not built in this tree.

This tab documents the **operational graph**: local loops, GitHub Actions, release tags, Docker images, Nuitka CLI/LSP binaries, Fernet-encrypted LSP metadata, GCP Cloud Build/Run, observability hooks, and security controls.

Conceptual model

```

flowchart TB
    subgraph Local
        DEV[make install / hatch]
        TEST[make test lint package]
        RUN[make run / run-lsp / docker-cli]
    end
    subgraph CI["GitHub Actions"]
        LINT[lint]
        PY[test matrix 3.10-3.13]
        PKG[sdist + wheel]
        DOCK[docker api + cli]
        VSIX[vscode-extension package]
    end
    subgraph Artifacts
        WHEEL[hoox-pyne wheel]
        CLIBIN[Nuitka pynescrypt CLI]
        LSPBIN[Nuitka pynescrypt-lsp]
        VSIXART[*vsix]
        IMG[GHCR api + cli + lsp]
    end
    subgraph Deploy
        CR[Cloud Run Pro API]
        PYPI[PyPI hoox-pyne]
        REL[GitHub Release]
    end
    DEV --> TEST --> LINT
    LINT --> PY --> PKG
    PY --> DOCK
    PKG --> WHEEL --> PYPI
    REL --> CLIBIN
    REL --> LSPBIN
    REL --> VSIXART
    IMG --> CR

```

Invariant: CI green on `main` is necessary but not sufficient for a full release. Tag `v*` fires **Build & Release** (binaries + VSIX + GitHub Release), **Publish** (`hoox-pyne` from `__about__.py`, currently 0.6.7, to PyPI), and **GHCR** (`ghcr.io/hoox-sh/pyne/{api,cli,lsp}`). Cloud Build deploys the API image separately.



Interface surface

Concern	Entry	Doc
Local install & loops	<code>Makefile</code> , hatch envs	Local development
PR / push CI	<code>.github/workflows/ci.yml</code> (name: CI)	CI
Versioned CLI + LSP + VSIX	<code>.github/workflows/release.yml</code> (name: Build & Release)	Release
PyPI (hoox-pyne)	<code>.github/workflows/publish.yml</code> (name: Publish)	PyPI publish
GHCR images	<code>.github/workflows/ghcr.yml</code> (name: GHCR)	Docker
First ship checklist	personal PyPI + API token	Publish checklist
Containers	<code>Dockerfile</code> targets <code>api/api-dev/lsp/cli</code>	Docker
Compiled CLI / LSP	<code>scripts/build/compile.py --target</code>	Nuitka build
Builtin metadata crypto	Fernet key + <code>.enc</code>	Metadata crypto
Cloud Run	<code>cloudbuild.yaml</code>	GCP
Logs / health	Flask / , gunicorn, Cloud Logging	Observability
Auth, CORS, secrets	<code>backend/middleware/auth.py</code>	Security

Internals (repo map)

Path	Role
<code>Makefile</code>	Human-facing orthography: install, test, lint, build, docker, worker
<code>pyproject.toml</code>	Hatch envs (<code>test</code> , <code>lint</code> , <code>docs</code>), optional extras (<code>lsp</code> , <code>data</code>)
<code>.github/workflows/</code>	<code>ci.yml</code> , <code>release.yml</code> , <code>publish.yml</code> , <code>ghcr.yml</code>
<code>scripts/build/</code>	Nuitka compile + CI build + Fernet metadata stage
<code>scripts/generate_builtin_metadata.py</code>	Regenerates LSP <code>builtin_metadata.json</code> from live builtins
<code>Dockerfile</code> / <code>docker-bake.hcl</code>	Multi-target Buildx images (api, api-dev, lsp)
<code>docker-compose.yml</code>	Local API (+ optional Redis / LSP profiles)
<code>cloudbuild.yaml</code>	Build → GCR push → Cloud Run deploy; optional LSP compile
<code>vscode-extension/</code>	Node 22 extension package / vsce

Invariants & edge cases

- Two console scripts, two entrypoints.** Preferred: `pyne` (Click) ≠ `pyne-lsp` (pygls). Aliases: `pynescrypt` / `pynescrypt-lsp`. Ops scripts must not conflate them.
- Generated artifacts are not hand-edited.** ANTLR/ASDL under `generated/` and `builtin_metadata.json` are code-derived.
- Fernet key is gitignored.** Without a stable `CRYPTO_KEY` / `METADATA_KEY` secret, every CI encrypt produces a different `.enc` blob (harmless functionally, bad for reproducibility).
- Python matrix vs Nuitka pin.** CI tests 3.10–3.13; Nuitka release currently pins **3.11** (`release.yml` `PYTHON_VERSION` , `nuitka>=2.5.1,<2.8`).



5. **AXIS CI is not in this repo.** Playwright / PWA security gates live in [hoox-sh/axis](#). There is no `axis-nightly.yml` here.

Worked examples

```
# Fast local confidence loop
make install
make lint
make test-lsp
make build-check    # import check only, ~30s, no Nuitka compile

# API in Docker – local *dev* stack (api-dev + source mounts)
make docker-up
make docker-smoke

# Production image bake (load local) + optional prod compose overlay
make docker-build
# export ADMIN_TOKEN=... && make docker-prod
```

Failure modes

Symptom	Likely cause	Fix
CI lint red, local green	Different ruff/mypy versions	Match CI install pins in workflow
Metadata decrypt fails in binary	Missing key at runtime	Set <code>PYNESCRIPT_METADATA_KEY</code> or embed key at build
Cloud Run 502 after deploy	Image missing deps / gunicorn bind	Confirm <code>Dockerfile</code> target <code>api</code> ENTRYPOINT + <code>PORT</code> / <code>GUNICORN_BIND</code>
VSIX empty / missing	<code>npm ci</code> / vsce not run	Use <code>make build-vscode</code> or release job artifacts
Nuitka Anaconda link error	Static libpython missing	<code>conda install libpython-static</code> or keep <code>--static-libpython=no</code>

See also

- [Contributing](#)
- [LSP architecture](#)
- [Pro API contract](#)
- [pyne-worker](#)
- [pyne-agent-worker](#)

LOCAL DEV

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title: "Local development" description: "Install, test, lint, package, and run PYNE — Make targets after AXIS extraction."

LOCAL DEVELOPMENT

Abstract

A PYNE checkout is a **Python-first workspace**: src-layout package + Flask Pro API, optional VS Code extension (Node). PyneTS is the standalone [hoox-sh/pynets](#) / [@hoox-sh/pynets](#) package. The AXIS charting PWA is a **sister repo** ([axis](#)) — do not recreate [frontend/](#) here.



Conceptual model

```
flowchart LR
  SRC[src/pynescrypt] --> CLI[pyne CLI]
  SRC --> LSP[pyne-lsp]
  SRC --> LIB[import pynescrypt]
  BE[backend/] --> API[Flask :5002]
  VSC[vscode-extension/] --> VSIX[vsce package]
  TS[hoox-sh/pynets] --> BUN[bun test]
  DOCK[Dockerfile] --> IMG[api · cli · lsp]
```



Interface surface

Make targets (primary)

Target	Effect
make install	pip install -e "[lsp,pro]"
make test	full pytest tests/
make test-cli	tests/test_cli.py
make test-lsp	langserver + lsp_features
make test-backend	tests/test_backend.py
make lint / make fmt	Ruff on src/ , tests/ , backend/
make package	sdist + wheel (python -m build)
make build	Nuitka LSP binary
make build-cli	Nuitka CLI binary
make build-check	import check for lsp+cli (~30s)
make build-vscode	compile + package VSIX
make run	Flask Pro API (:5002 typical)
make run-lsp	python -m pynescrypt.langserver
make docker-build	bake api
make docker-build-cli	bake cli
make docker-build-lsp	bake lsp (pynescrypt-lsp:latest ; GHCR name ghcr.io/hoox-sh/pyne/lsp)
make docker-build-all	api + api-dev + lsp + cli
make docker-buildx	multi-platform release bake (api + cli + lsp)
make docker-push-ghcr	gh workflow run ghcr.yml
make docker-cli ARGS="..."	compose profile cli (ephemeral)
make deploy-vps	rsync + restart API; health API :5002, AXIS :80 (VPS; local AXIS is :8081)
make docker-up	compose API on :5002
make docker-up-full	+ redis profile
make docker-prod	gunicorn overlay (ADMIN_TOKEN required)
make docker-smoke	curl health on :5002
make docker-down / docker-logs	stop / logs



Hatch

```
hatch run test:test
hatch run test:test-cov
hatch run lint:style
hatch run lint:typing
hatch run lint:gen-parser
```

Quick desk loop

```
git clone https://github.com/hoox-sh/pyne.git
cd pyne
python3 -m venv .venv && source .venv/bin/activate
pip install -e ".[lsp,compile,pro]"
pyne info
pyne check examples/rsi_strategy.pine
make test-cli
make run    # Pro API
```

From PyPI (no checkout)

```
pip install "hoox-pyne[lsp,compile]"
pyne --version    # 0.6.7
```

Internals

Path	Why it matters
src/pynescrypt/	Editable package root
tests/conftest.py	Optional <code>--example-scripts-dir</code> only; no third-party corpus shipped
backend/requirements.txt	Pro API deps (also covered by <code>[pro]</code> extra)
hoox-sh/pynets	Standalone TS library (<code>@hoox-sh/pynets</code>); not an in-tree submodule
vscode-extension/	PYNE VS Code extension
Dockerfile	targets <code>api</code> , <code>api-dev</code> , <code>lsp</code> , <code>cli</code>

Python

- **Requires-Python:** `>=3.10`
- **CI matrix:** 3.10–3.13
- **Hatch test matrix:** 3.10–3.12 (no 3.13 cell)
- **Recommended:** 3.12+ with `venv`



Optional extras

Extra	Adds
lsp	pygls / lsprotocol → pyne-lsp (alias pynescrypt-lsp)
compile	numpy / numba
data / datafeed	ccxt
pro	Flask + gunicorn + matplotlib + redis + compile

Invariants

1. No `frontend/` in this repo — AXIS is external.
2. `from __future__ import annotations` required on new Python modules.
3. **Grammar edits** only under `resource/`; regenerate via hatch, don't hand-edit generated parser.
4. **Corpus fixture** runs hundreds of cases — prefer focused unit tests for CI paths.
5. **CLI Docker** runs as uid 1000; bind-mounts must be readable by others.

Failure modes

Symptom	Fix
No module named pynescrypt	activate venv; <code>pip install -e .</code> or <code>hoox-pyne</code>
Pro API ImportError flask	<code>pip install "hoox-pyne[pro]"</code> or backend requirements
Nuitka missing	<code>pip install nuitka</code> ; <code>make build-check</code> first
Docker CLI can't read file	<code>chmod</code> host path; don't use 0700-only dirs

See also

- [Installation](#)
- [CI](#)
- [Docker](#)
- [Nuitka build](#)

CI

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title: "Continuous integration" description: "GitHub Actions for lint, Python 3.10–3.13 tests, package build, Docker api+cli smoke, and VS Code extension."

CONTINUOUS INTEGRATION

Abstract

CI is the **public invariant check** for the **pyne** repo after AXIS extraction. Workflows live under `.github/workflows/`. AXIS PWA/e2e CI lives in the sister **axis** repo — **not** here.



Workflow file	name :	Role
ci.yml	CI	Lint, test matrix, package, Docker api+cli, VSIX
release.yml	Build & Release	Tag v* + dispatch: wheels, CLI/LSP Nuitka binaries, CLI image tarball, VSIX, GitHub Release
publish.yml	Publish	Tag v* + dispatch: PyPI hoox-pyne (token or OIDC)
ghcr.yml	GHCR	Tag v* + dispatch: ghcr.io/hoox-sh/pyne/{api,cli,lsp} multi-arch
docs-versions.yml	Docs versions	Daily: refresh PyPI / npm / Open VSX / GitHub release stamps. Commits only on change. No docs pageload.

Conceptual model

```
flowchart TB
    PUSH[push / PR → main] --> CONC[concurrency: ci-ref]
    CONC --> LINT[lint]
    CONC --> TEST[test matrix 3.10-3.13]
    CONC --> PKG[package sdist+wheel]
    CONC --> DOCK[docker api + cli]
    CONC --> VSC[vscode-ext-test]
    TAG[tag v*] --> REL[release.yml Build & Release]
    TAG --> PUB[publish.yml Publish]
    TAG --> GHCR[ghcr.yml GHCR]
```

Interface surface

Triggers

Workflow	On	Purpose
CI	push/PR to main	Required confidence
Build & Release	tags v* + dispatch	Binaries + VSIX + GitHub Release
Publish	tags v* + dispatch (dry_run default true)	PyPI hoox-pyne
GHCR	tags v* + dispatch	Public images api / cli / lsp



ci.yml jobs (current)

Job	Runtime	What it asserts
lint	Python 3.13	Ruff E/F/W gate on <code>src/</code> <code>tests/</code> (+ auth middleware); mypy non-blocking
test	Matrix 3.10–3.13	<code>. [lsp,pro,compile]</code> + backend deps; core (<code>test_linter</code> / <code>test_evaluator</code> / <code>test_cli</code>); runtime (parity, strategy, series, incremental TA, package Runtime); LSP (<code>test_langserver</code> + <code>test_lsp_features</code>); backend; Codecov on 3.13
package	3.13	<code>python -m build</code> + twine; wheel smoke (<code>pynescrypt --help</code>); artifact
docker	buildx	Build <code>api</code> + <code>cli</code> targets; API health + admin fail-closed; CLI <code>--help</code> / <code>info</code> / <code>check</code> (<code>--network=None</code>)
vscode-ext-test	Node 22	<code>npm ci</code> → compile → <code>vsce package</code> → VSIX artifact

Concurrency: `group: ci-${{ github.ref }}`, `cancel-in-progress: true`. Default permissions: `contents: read`.

Not in this repo's CI

- AXIS PWA / Playwright (axis repo — no `axis-nightly.yml` here)
- GHCR push (separate `ghcr.yml`, tag-triggered)
- Cloudflare Worker deploy
- Full `tests/` corpus parametrized suite (run locally: `make test`)

Internals

Path	Role
<code>.github/workflows/ci.yml</code>	PR/main fan-out (<code>name: CI</code>)
<code>.github/workflows/release.yml</code>	Multi-artifact tag release (<code>name: Build & Release</code>)
<code>.github/workflows/publish.yml</code>	PyPI (token preferred) (<code>name: Publish</code>)
<code>.github/workflows/ghcr.yml</code>	Multi-arch image push (<code>name: GHCR</code>)
<code>tests/test_cli.py</code>	CLI unit suite (CI-critical)

Secrets

Secret	Used by	Notes
<code>PYPI_API_TOKEN</code>	<code>publish.yml</code>	Personal PyPI account (<code>jango-blockchained</code>)
<code>METADATA_KEY</code> / <code>CRYPTO_KEY</code>	<code>release.yml</code> LSP encrypt	Fernet metadata
<code>VSCE_PAT</code>	optional Marketplace	VSIX still attaches without it
<code>GITHUB_TOKEN</code>	release upload	default

Invariants

1. **Fail-fast is off** on the Python matrix.



- 2. **Codecov only on 3.13.**
- 3. **CLI Docker image is non-root** — smoke mounts must be world-readable (`chmod` on `mktemp`).
- 4. **Corpus parametrized tests are intentionally skipped** in CI for time; local `make test` is broader.
- 5. **Package smoke** installs the wheel and runs `pynescrypt --help` .

Worked examples

```
# Lint as CI (correctness gate)
ruff check src/ tests/ backend/middleware/auth.py --select E,F,W --ignore E501,E741,F841,W293,E402

# Core unit slice
pip install -e ".[lsp,pro,compile]"
python -m pytest tests/test_linter.py tests/test_evaluator.py tests/test_cli.py -q

# Docker CLI smoke
docker buildx bake cli
docker run --rm --network=none pynescrypt-cli:latest info
```

Failure modes

Symptom	Cause	Fix
Docker CLI <code>PermissionError</code> on <code>smoke.pine</code>	<code>mktemp</code> 0700 + non-root user	CI <code>chmods</code> 755/644 (already fixed)
Wheel missing <code>pynescrypt</code>	wrong package name	install <code>hoox-pyne</code> / built wheel
Ruff F821 on main	incomplete imports	fix before push; gate is E,F,W

See also

- [Release](#)
- [PyPI publish](#)
- [Docker](#)

RELEASE

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title: "Release" description: "Tag-driven CLI + LSP multi-platform Nuitka binaries, wheels, Docker CLI image, VSIX packaging, GitHub Releases, and Marketplace publish."

RELEASE

Abstract

A PYNE release is a **multi-artifact event**, not a single wheel upload. The primary automation (`.github/workflows/release.yml`) builds:

- **PyPI sdist + wheel** (`hoox-pyne`) with CLI smoke
- **CLI platform-native Nuitka binaries** (`pynescrypt`)



- **LSP** platform-native Nuitka binaries (`pynescrypt-lsp`)
- **CLI Docker image** tarball (`pynescrypt-cli-image.tar.gz`)
- **VS Code** VSIX

Assets attach to a GitHub Release; PyPI upload is handled by **Publish** (`publish.yml`). Multi-arch container images are a **separate** workflow (**GHCR** / `ghcr.yml`) — not attached as tarballs except the CLI image save in this workflow. Marketplace / Open VSX publish is optional via `VSCE_PAT` / `OVSX_PAT` .

Conceptual model

```
sequenceDiagram
    participant Dev as Maintainer
    participant GH as GitHub
    participant Pkg as build-package
    participant CLI as build-cli matrix
    participant LSP as build-lsp matrix
    participant Dock as build-docker-cli
    participant VSC as build-vscode
    participant Rel as release job
    participant Mkt as Marketplace
    Dev->>GH: git tag vX.Y.Z && push
    GH->>Pkg: sdist + wheel
    GH->>CLI: linux / windows / macos
    GH->>LSP: linux / windows / macos
    GH->>Dock: pynescrypt-cli image
    GH->>VSC: Node 22 vsce package
    Pkg->>Rel: wheels
    CLI->>Rel: CLI binaries
    LSP->>Rel: LSP binaries
    Dock->>Rel: image tarball
    VSC->>Rel: VSIX
    Rel->>GH: softprops/action-gh-release
    VSC->>Mkt: vsce publish (on tag)
```

Interface surface

Triggers

Event	Behavior
Push tag matching <code>v*</code>	Full build + create release + marketplace/Open VSX publish
<code>workflow_dispatch</code> with <code>version</code> input	Same jobs; Create Release also runs on dispatch (not tag-only)



Jobs

Job	Matrix / OS	Output
build-package	ubuntu-latest	hoox-pyne-dist (sdist + wheel)
build-cli	ubuntu-latest, windows-latest, macos-latest	pynescrypt-cli-linux-x86_64 , ...-windows-x86_64.exe , ...-macos-arm64
build-lsp	ubuntu-latest, windows-latest, macos-latest	pynescrypt-lsp-linux-x86_64 , ...-windows-x86_64.exe , ...-macos-arm64
build-docker-cli	ubuntu-latest	pynescrypt-cli-docker (gzipped docker save)
build-vscode	ubuntu-latest	pynescrypt-vscode-extension VSIX
release	needs package + vscode success; tag or dispatch	GitHub Release body + staged assets (name: pyne vX.Y.Z)
publish-vscode	needs VSIX; tag or dispatch	vsce publish if VSCE_PAT ; ovsx publish if OVSX_PAT

Environment

```
env:  
  PYTHON_VERSION: "3.11"    # Nuitka 2.5-2.7 graph; comment in workflow  
  NUITKA_JOBS: 4
```

Build steps rely on:

```
# LSP  
pip install -e ".[lsp]" "nuitka>=2.5.1,<2.8" cryptography  
python scripts/build/compile.py --target lsp --no-encrypt --check  
python scripts/build/ci_build.py --target lsp --jobs 4  
# CRYPTO_KEY from secrets.METADATA_KEY  
  
# CLI  
pip install -e . "nuitka>=2.5.1,<2.8"  
python scripts/build/compile.py --target cli --check  
python scripts/build/ci_build.py --target cli --jobs 4 --skip-metadata --skip-vsix
```

Local make-side packaging

```
make package      # sdist + wheel (python -m build)  
make build        # Nuitka LSP  
make build-cli    # Nuitka CLI  
make build-vscode # npm install && compile && vsce package  
make docker-build-cli
```

Artifacts land under `dist/` , `dist/lsp/` , `dist/cli/` , `dist/vsrx/` , and `vscode-extension/*.vsrx` depending on script path.



Internals

Path	Role
<code>.github/workflows/release.yml</code>	Orchestration
<code>scripts/build/ci_build.py</code>	CI-oriented Nuitka + metadata encrypt
<code>scripts/build/compile.py</code>	Local/full compile options (<code>--onefile</code> , <code>--standalone</code> , <code>--check</code>)
<code>vscode-extension/package.json</code>	Extension version / engines
<code>src/pynescrypt/__about__.py</code>	Hatch dynamic version for Python package

Release asset contract (from release body)

Artifact	Install sketch
<code>hoox_pyne-*.whl</code> / <code>*.tar.gz</code>	<code>pip install ./hoox_pyne-*.whl</code>
<code>pynescrypt-cli-linux-x86_64</code>	<code>chmod +x</code> → <code>/usr/local/bin/pynescrypt</code>
<code>pynescrypt-cli-windows-x86_64.exe</code>	Place on PATH as <code>pynescrypt.exe</code>
<code>pynescrypt-cli-macos-arm64</code>	Place on PATH (Apple Silicon runners)
<code>pynescrypt-cli-image.tar.gz</code>	<code>gunzip -c ... docker load</code> → <code>pynescrypt-cli:latest</code>
<code>pynescrypt-lsp-linux-x86_64</code>	<code>chmod +x</code> → <code>/usr/local/bin/pynescrypt-lsp</code>
<code>pynescrypt-lsp-windows-x86_64.exe</code>	Place on PATH
<code>pynescrypt-lsp-macos-arm64</code>	Place on PATH (Apple Silicon runners)
<code>pyne-vscode-*.vsix</code> (<code>hoox-sh.pyne 0.4.4</code>)	<code>code --install-extension pyne-vscode-*.vsix</code> · site: hoox.sh/pyne-vscode.vsix (unversioned)

Invariants & edge cases

1. **CRYPTO_KEY** must be stable across builds if you care about byte-identical `builtin_metadata.json.enc` .
Supply `secrets.METADATA_KEY` .
2. **Tag name is the version source** for the GitHub Release title (`v` prefix stripped).
3. **Marketplace publish needs VSCE_PAT** . Without it, packaging may still succeed while publish fails.
4. **Artifact retention is short** (5 days on build jobs) — releases must attach files immediately.
5. **macOS artifact is ARM64** from `macos-latest` — Intel Mac users may need separate builds if required.
6. **Draft vs prerelease**: workflow currently sets `draft: false` , `prerelease: false` for tag releases.



Worked examples

Cut a release

```
# ensure main is green
git checkout main && git pull
# bump versions in extension / __about__ as needed
git tag -a v0.6.7 -m "v0.6.7"
git push origin v0.6.7
# watch Actions → Build & Release, Publish, GHCR
```

Smoke-test a downloaded binary

```
chmod +x pynescrypt-cli-linux-x86_64
./pynescrypt-cli-linux-x86_64 --help
./pynescrypt-cli-linux-x86_64 check script.pine

chmod +x pynescrypt-lsp-linux-x86_64
./pynescrypt-lsp-linux-x86_64 --help
# or wire stdio into an editor client
```

Failure modes

Symptom	Cause	Fix
Binary not found after Nuitka	Path layout drift in <code>ci_build.py</code>	Inspect <code>dist/</code> ; fix finder step
Decrypt errors in field	Key mismatch between encrypt and runtime	Align <code>METADATA_KEY</code> with embedded key strategy
Empty GitHub Release assets	Artifact download path mismatch	Match <code>files:</code> globs to download-artifact layout
Marketplace 401	Bad/expired <code>VSCE_PAT</code>	Rotate PAT with Marketplace scopes
Windows <code>.exe</code> not marked executable	Finder uses <code>-type f -executable</code>	Fallback non-executable find branch in workflow

See also

- [Nuitka build](#)
- [Metadata crypto](#)
- [CI](#)
- [VS Code extension](#)

PYPI PUBLISH

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title: "PyPI publish" description: "Ship hoox-pyne to PyPI under the personal jango-blockchained account (API token or Trusted Publishing). Import package stays pynescrypt."

PYPI PUBLISH

Abstract

The Python distribution name on PyPI is `hoox-pyne`. After install, the import path stays `pynescrypt`; preferred CLIs are `pyne` / `pyne-lsp` (aliases: `pynescrypt` / `pynescrypt-lsp`).

PyPI ownership is the personal account `jango-blockchained` (not a PyPI organization — org approval is not required). Automation still runs on GitHub `hoox-sh/pyne` via `.github/workflows/publish.yml`.



Auth modes (first match wins):

1. **API token** — repo secret `PYPI_API_TOKEN` from the personal PyPI account (recommended while no PyPI org is needed).
2. **Trusted Publishing (OIDC)** — pending publisher created while logged in as `jango-blockchained`, with GitHub Owner = `hoox-sh` (where Actions run).

Why not `pynescrypt` or `pyne` on PyPI? `pynescrypt` is owned by `elbakramer/pynescrypt`; plain `pyne` / `PyNE` is an unrelated 0.1.0 process-networking library. We ship as `hoox-pyne`; the import package stays `pynescrypt`.

Install surface

```
pip install hoox-pyne
pip install "hoox-pyne[lsp]"
pip install "hoox-pyne[pro]"

python -c "import pynescrypt; print(pynescrypt.__version__)"
pyne --help
pyne-lsp # stdio language server (alias: pynescrypt-lsp)
```

One-time setup (personal PyPI account)

A. Recommended — API token (no org, no OIDC)

1. Sign in at pypi.org as `jango-blockchained` (2FA required).
2. **Account settings** → **API tokens** → **Add API token**.
 - o Name: e.g. `hoox-sh-pyne-gha`
 - o Scope: **Entire account** for the first upload of `hoox-pyne`; after the project exists you can rotate to a **project-scoped** token.
3. Copy the token (`pypi-...`) once.
4. On GitHub:

```
# Environment (optional reviewers / wait timer)
gh api -X PUT repos/hoox-sh/pyne/environments/pypi

# Never commit the token — Actions secret only
gh secret set PYPI_API_TOKEN -R hoox-sh/pyne
# paste pypi-... token
```

5. First successful tag (or `workflow_dispatch` with `dry_run=false`) creates `hoox-pyne` under your personal account.

B. Optional — Trusted Publishing (OIDC)

Use this when you prefer no long-lived token. Log into PyPI as `jango-blockchained` (personal user), then:



Field	Value
PyPI project name	hoox-pyne
Owner	hoox-sh ← GitHub repo owner (workflow host), not your PyPI username
Repository	pyne
Workflow name	publish.yml
Environment name	pypi

Leave `PYPI_API_TOKEN` **unset** so the publish job uses OIDC.

Do **not** put `jango-blockchained` in the GitHub Owner field unless the workflow actually runs under `github.com/jango-blockchained/pyne`. OIDC is bound to the repo that executes the job (`hoox-sh/pyne`).

GitHub environment

Repo **Settings** → **Environments** → `pypi` on `hoox-sh/pyne`. Optional: required reviewers on production uploads.

Cut a release

```
# 1. Version + changelog on main
#   src/pynascript/__about__.py → 0.6.7
#   vscode-extension/package.json is a separate pin (currently 0.4.4)
#   CHANGELOG.md

# 2. Local smoke (no upload)
pip install build twine
rm -rf dist/
python -m build
twine check dist/*
# expect: hoox_pyne-0.6.7-*.whl and hoox_pyne-0.6.7.tar.gz

# 3. Push main, then tag
git tag -a v0.6.7 -m "v0.6.7"
git push origin v0.6.7

# 4. Watch Actions → Publish
```

If a tag already exists and the wheel was never uploaded, re-run:

Actions → **Publish** → **Run workflow** → `dry_run=false` (builds from that workflow's default branch / checked-out tag context as configured), or retag a new patch version.

Dry-run without upload: **Actions** → **Publish** → **Run workflow** → `dry_run=true`.

Workflow map

Job	When	What
build	tag <code>v*</code> or dispatch	<code>python -m build</code> + <code>twine check</code> + artifact
publish-pypi	tag <code>v*</code> or dispatch with <code>dry_run=false</code>	Token if <code>PYPI_API_TOKEN</code> set, else OIDC



Failure modes

Symptom	Cause	Fix
403 invalid token	Wrong/revoked <code>PYPI_API_TOKEN</code>	Create new token as <code>jango-blockchained</code> ; re- <code>gh secret set</code>
403 / no trusted publisher	OIDC path without pending publisher	Add pending publisher or set API token
Wrong GitHub Owner in pending publisher	Used PyPI username as Owner	Set Owner to <code>hoox-sh</code> (GitHub org of the repo)
Environment not found	Missing GitHub <code>pypi</code> env	<code>gh api -X PUT repos/hoox-sh/pyne/environments/pypi</code>
File already exists	Version already on PyPI	Bump <code>__about__.__version__</code> and tag a new version
Tag publish skipped	<code>dry_run</code> dispatch only	Use a real <code>v*</code> tag or <code>dry_run=false</code>

See also

- [Release](#) — CLI/LSP binaries + VSIX
- [Publish checklist](#)
- [CI](#)
- Root `CONTRIBUTING.md`

PUBLISH CHECKLIST

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title: "Publish checklist" description: "Publish hoox-pyne under personal PyPI account jango-blockchained from hoox-sh/pyne Actions."

PUBLISH CHECKLIST

Abstract

Ship `hoox-pyne` from GitHub `hoox-sh/pyne` while owning the PyPI project under the **personal** account `jango-blockchained`. A PyPI **organization is not required** (and is not used while org approval is pending).



0. Identity map

Surface	Value
GitHub org / repo	hoox-sh/pyne (Actions host)
PyPI account (owner)	jango-blockchained (personal)
PyPI project	hoox-pyne
Import / CLIs	import pynescrypt ; CLIs pyne / pyne-lsp (aliases pynescrypt / pynescrypt-lsp)
Package version	src/pynescrypt/__about__.py → 0.6.7
VS Marketplace / Open VSX publisher	hoox-sh (hoox-sh.pyne 0.4.4)

1. Confirm repo host (hoox-sh/pyne)

The GitHub repo **already lives** at `hoox-sh/pyne` . If a checkout still points at the old personal remote, retarget it:

```
git remote set-url origin https://github.com/hoox-sh/pyne.git
git remote -v
gh repo view --json nameWithOwner,url
# expect: hoox-sh/pyne
```

Automation secrets

Item	Action
Actions enabled	Org/repo Settings → Actions allowed for this repo
Environment <code>pypi</code>	Recreate if missing: <code>gh api -X PUT repos/hoox-sh/pyne/environments/pypi</code>
Secrets	Re-set if missing: <code>METADATA_KEY</code> , <code>CRYPTO_KEY</code> (same Fernet material as <code>scripts/build/.metadata.key</code>)
Optional	<code>VSCE_PAT</code> for Marketplace publish on tag
Workflow permissions	Read/write as needed for <code>release.yml</code> (contents: write on release job)

```
gh secret set METADATA_KEY -R hoox-sh/pyne < scripts/build/.metadata.key
gh secret set CRYPTO_KEY -R hoox-sh/pyne < scripts/build/.metadata.key
# optional marketplace:
# gh secret set VSCE_PAT -R hoox-sh/pyne
```

2. PyPI under personal account `jango-blockchained`

Preferred — API token

1. pypi.org as `jango-blockchained` → **API tokens** → add token.
2. `gh secret set PYPI_API_TOKEN -R hoox-sh/pyne`
3. First upload creates project `hoox-pyne` owned by your user.



Optional — Trusted Publishing

Log in as `jango-blockchained` (personal), then pending publisher:

Field	Value
PyPI project name	<code>hoox-pyne</code>
Owner	<code>hoox-sh</code> (GitHub repo owner — Actions host)
Repository	<code>pyne</code>
Workflow name	<code>publish.yml</code>
Environment name	<code>pypi</code>

Leave `PYPI_API_TOKEN` unset to force OIDC. Do **not** put your PyPI username in the GitHub Owner field.

Details: [PyPI publish](#).

3. Local package smoke (no upload)

```
# version already 0.6.7 in src/pynescrypt/__about__.py
pip install build twine
rm -rf dist/
python -m build
twine check dist/*
# expect: hoox_pyne-0.6.7-py3-none-any.whl hoox_pyne-0.6.7.tar.gz

python -m venv /tmp/pyne-smoke
/tmp/pyne-smoke/bin/pip install dist/*.whl
/tmp/pyne-smoke/bin/python -c "import pynescrypt; print(pynescrypt.__version__)"
/tmp/pyne-smoke/bin/pyne --help
```

4. GitHub Actions dry-run

1. Push `main` with publish-ready tree to `hoox-sh/pyne`.
2. Confirm **CI** green (lint, test matrix, package, docker, vscode-ext).
3. Actions → **Publish** → Run workflow → `dry_run=true` (build + twine only, no upload).

5. Cut `v0.6.7`

1. Move `CHANGELOG.md` **Unreleased** notes into `### [0.6.7] - YYYY-MM-DD`.
2. Align VSIX version in `vscode-extension/package.json` (already **0.4.4**) if shipping Marketplace/VSIX with this tag.
3. Tag and push:

```
git tag -a v0.6.7 -m "v0.6.7"
git push origin v0.6.7
```

4. Watch in parallel:

- **Publish** → PyPI `hoox-pyne`
- **Build & Release** → GitHub Release assets + optional Marketplace



- **GHCR** → `ghcr.io/hoox-sh/pyne/{api,cli,lsp}:0.6.7`

6. Post-publish verify

```
pip index versions hoox-pyne # or: pip install hoox-pyne==0.6.7
python -c "import pynescrypt; print(pynescrypt.__version__)"
# GitHub Release: https://github.com/hoox-sh/pyne/releases/tag/v0.6.7
# PyPI: https://pypi.org/project/hoox-pyne/0.6.7/
```

Hardcoded owner map (repo automation)

Workflows use relative `github.*` context (no hard-coded owner) except documentation/comments. Project metadata and docs must point at **hoox-sh/pyne** :

Location	Role
<code>pyproject.toml</code> <code>[project.urls]</code>	PyPI package links
<code>.github/workflows/publish.yml</code> header	Trusted Publisher fields
<code>Dockerfile</code> <code>org.opencontainers.image.source</code>	Image source label
<code>package.json</code> / <code>vscode-extension/package.json</code> repository	Clone / issues URLs
<code>docs/pyne/docs.json</code> navbar GitHub	Docs site link
<code>scripts/build/README.md</code> <code>gh secret set -R</code>	Secret target repo
<code>CONTRIBUTING.md</code> + this page	Human publish runbook

Intentionally not changed

Item	Why
<code>.github/FUNDING.yml</code>	GitHub Sponsors username (personal)
Sister repos <code>jango-blockchained/{axis,hoox}</code>	Separate transfers
Author strings <code>jango_blockchained</code>	Copyright / author field
PyPI account <code>jango-blockchained</code>	Package owner; not the VS Code namespace

See also

- [PyPI publish](#)
- [Release](#) — LSP + VSIX
- [CI](#)
- Root `CONTRIBUTING.md`

DOCKER

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title: "Docker" description: "Buildx multi-target images, Compose profiles, and Cloud Run production contract for the PYNE Pro API."

DOCKER

Abstract

Containers package the **Pro API** (Flask + evaluator), the **CLI** (`pynescrypt`), and the **LSP** for reproducible local runs and Cloud Run deploys. A single multi-stage `Dockerfile` exposes named targets; **Docker Buildx Bake** (`docker-bake.hcl`) drives local loads and multi-platform release builds. Compose wires volume mounts and optional Redis/LSP/CLI profiles for desk development, with a production overlay for gunicorn.



Conceptual model

```
flowchart LR
    subgraph Bake
        B[docker buildx bake]
    end
    subgraph Targets
        T1[api gunicorn]
        T2[api-dev Flask]
        T3[lsp stdio]
        T4[cli pyonescript]
    end
    subgraph Runtime
        CR[Cloud Run :8080]
        CMP[compose :5002]
        CLIRUN[docker run cli]
    end
    B --> T1
    B --> T2
    B --> T3
    B --> T4
    T1 --> CR
    T2 --> CMP
    T4 --> CLIRUN
    COMPOSE[docker-compose.yml] --> T2
    PROD[docker-compose.prod.yml] --> T1
```

Interface surface

Dockerfile targets

Target	Process model	Audience
api	entrypoint-api.sh → gunicorn (0.0.0.0:\$PORT)	Production / Cloud Run / compose prod
api-dev	python -m backend.app with HOST=0.0.0.0	Local compose (source mounts)
lsp	ENTRYPOINT pyne-lsp (stdio)	Compose profile lsp ; GHCR ghcr.io/hoox-sh/pyne/lsp
cli	entrypoint-cli.sh → pyne (fallback pyonescript ; ephemeral; /work CWD)	CI gates, one-shot parse/lint/compile/run

All targets: Python 3.12-slim, non-root `appuser` (uid 1000). API targets also set `API_KEY_STORE=/data/api_keys.json` , `MPLBACKEND=Agg` , healthcheck via `curl` . The `cli` image installs `.` `[compile,data]` only (no Flask/matplotlib) and is smaller/faster to build than `api` .



Buildx Bake (`docker-bake.hcl`)

```
# Local load (host platform)
docker buildx bake api          # production image → pynescrypt-api:latest
docker buildx bake cli          # CLI image → pynescrypt-cli:latest (ENTRYPOINT: pyne)
docker buildx bake lsp          # LSP image → pynescrypt-lsp:latest (ENTRYPOINT: pyne-lsp)
docker buildx bake              # default group: api + api-dev
docker buildx bake all          # api + api-dev + lsp + cli

# Multi-platform (amd64 + arm64). Set REGISTRY to push:
REGISTRY=ghcr.io/hoox-sh/pyne TAG=0.6.7 docker buildx bake release
# release group = api-release + cli-release + lsp-release
# → ghcr.io/hoox-sh/pyne/{api,cli,lsp}:0.6.7
# Prefer: make docker-push-ghcr (workflow GHCR on v* tags)
```

Make wrappers:

```
make docker-build      # bake api
make docker-build-cli  # bake cli
make docker-build-lsp  # bake lsp
make docker-build-all # bake all targets
make docker-buildx     # bake release (multi-platform)
make docker-push-ghcr  # gh workflow run ghcr.yml
make docker-cli ARGS="check script.pine" # compose profile cli
```

One-time multi-platform builder (if needed):

```
docker buildx create --use --name pynescrypt
```

Compose

```
cp .env.example .env # optional overrides

make docker-up        # api-dev on host :5002
make docker-up-full   # + redis profile (api + redis)
make docker-prod       # requires ADMIN_TOKEN; gunicorn, no source mounts
make docker-smoke      # curl http://127.0.0.1:5002/
make docker-down

# stdio LSP (not a long-running compose service)
docker compose --profile lsp run --rm lsp

# CLI (ephemeral; mounts repo at /work)
docker compose --profile cli run --rm cli check script.pine
# ≡ make docker-cli ARGS="check script.pine"
```

Service	Profile	Notes
api	default	Port <code>\${API_PORT:-5002}</code> :8080, no mounts of <code>src/</code> + <code>backend/</code> , volume <code>api_data</code> → <code>/data</code>
redis	redis	redis:7-alpine, AOF, <code>\${REDIS_PORT:-6379}</code>
lsp	lsp	stdio; use <code>docker compose run --rm lsp</code> (not detached <code>up</code>)
cli	cli	<code>entrypoint-cli.sh</code> → <code>pyne</code> ; mounts <code>./work</code> + <code>src</code> + <code>cli_data</code>



Network name: `pynescrypt-dev`.

Production overlay (`docker-compose.prod.yml`): target `api`, **volumes: !override** so only `api_data:/data` remains (dev source binds are dropped), SQLite key store on `/data`, mem/cpu caps, requires `ADMIN_TOKEN`.

Environment variables

Variable	Default	Meaning
<code>PORT</code>	<code>8080</code>	Listen port inside the container
<code>HOST</code>	<code>0.0.0.0</code>	Bind address (dev Flask runner)
<code>API_PORT</code>	<code>5002</code>	Host port published to container <code>8080</code>
<code>FLASK_ENV</code>	<code>production</code> / <code>development</code>	App mode by target
<code>STORE_BACKEND</code>	<code>json</code> (prod overlay: <code>sqlite</code>)	<code>json</code> <code>sqlite</code> <code>redis</code>
<code>API_KEY_STORE</code>	<code>/data/api_keys.json</code>	JSON key store path (<code>STORE_BACKEND=json</code>)
<code>API_KEY_STORE_SQLITE</code>	<code>/data/api_keys.db</code>	SQLite path (<code>STORE_BACKEND=sqlite</code>)
<code>ALLOWED_ORIGINS</code>	compose default includes <code>https://hoox.sh</code> , <code>https://hoox.sh/axis</code> , and local AXIS :8081	CORS allow-list. VPS AXIS is :80 — do not copy 8081 into VPS env
<code>ADMIN_TOKEN</code>	unset	Required for <code>POST /auth/create_key</code> (<code>X-Admin-Token</code>) and prod compose
<code>PYNE_RUNNER</code>	<code>0</code>	Optional hosted runner (docs). <code>1</code> enables <code>/scripts</code> + <code>/cron/*</code>
<code>PYNE_RUNNER_SCHEDULER</code>	<code>0</code>	In-process bar-close poll. Prefer <code>GUNICORN_WORKERS=1</code> when on
<code>PYNE_RUNNER_DB</code>	<code>/data/runner.db</code>	SQLite on the <code>api_data</code> volume
<code>REDIS_URL</code>	empty in prod; compose dev default <code>redis://redis:6379/0</code>	Required when <code>STORE_BACKEND=redis</code>
<code>GUNICORN_WORKERS</code> / <code>GUNICORN_THREADS</code> / <code>GUNICORN_TIMEOUT</code>	<code>2</code> / <code>4</code> / <code>120</code> (entrypoint default; prod compose overlay defaults to <code>60</code>)	Prod entrypoint knobs
<code>GUNICORN_BIND</code>	<code>0.0.0.0:\${PORT}</code>	Optional gunicorn bind override

Published image names

GitHub Container Registry (workflow **GHCR**, tags `v*`):

```
ghcr.io/hoox-sh/pyne/api:0.6.7
ghcr.io/hoox-sh/pyne/cli:0.6.7
ghcr.io/hoox-sh/pyne/lsp:0.6.7
```

Cloud Build (`cloudbuild.yaml`, **--target api** only):



```
gcr.io/$PROJECT_ID/pynescrypt/pynescrypt-pro-api:$COMMIT_SHA
gcr.io/$PROJECT_ID/pynescrypt/pynescrypt-pro-api:latest
```

Substitution `_PYNESCRIPT_VERSION` in that YAML defaults to **0.6.7**, matching `__about__.py`.

Internals

Build stages

1. **base-os / base** — slim image, non-root `appuser`; API/LSP `base` also installs `curl` + freetype/png
2. **builder** — `build-essential`, pip cache mounts, install `backend/requirements.txt` then `pip install ".[lsp]"` into `/install`
3. **lsp-builder** — `pip install ".[lsp]"` only (no Flask / matplotlib)
4. **cli-builder** — `pip install ".[compile,data]"` only (no Flask stack)
5. **runtime / api / api-dev / lsp / cli** — final CMDs / ENTRYPOINTS and healthchecks (`lsp` has none — `stdio`)

`.dockerignore`

Keeps context small by excluding `tests/` (corpus), `docs/`, `brand/`, node modules, compiler `*.nbc` / `*.nbi` caches, `venvs`, and IDE cruft.

Compose healthchecks

API: `curl -fsS http://127.0.0.1:8080/`. Redis: `redis-cli ping`.

Invariants & edge cases

1. **Host port 5002 vs container 8080**. Local compose publishes API on **:5002**. Cloud Run uses **:8080**. VPS deploy (`scripts/deploy_vps.sh`) health-checks **API :5002** and **AXIS :80**. Local AXIS PWA is **:8081** — do not treat 8081 as the VPS PWA port.
2. **Read-only mounts in compose** mean dependency changes need image rebuild; source edits are visible via `PYTHONPATH` precedence.
3. **Production target is immutable** — no live mounts; use the prod overlay or Cloud Run.
4. **Non-root appuser** cannot write arbitrary paths; keep `API_KEY_STORE` under `/data`.
5. **Cloud Build always builds --target api** (gunicorn). Do not point it at `api-dev`.
6. **LSP is stdio** — not an HTTP service; use `docker compose run --rm lsp` rather than expecting a published port.
7. **CLI is ephemeral** — `entrypoint-cli.sh` execs `pyne` (fallback `pynescrypt`); pass subcommands after the image name. Workdir defaults to `/work` (compose mounts the repo there).
8. **Prod overlay is fail-closed** — `ADMIN_TOKEN` is required (`:?` interpolation); `volumes: !override` drops the dev source binds so only `api_data:/data` remains.



Worked examples

Production-like local API (bake)

```
docker buildx bake api
docker run --rm -p 8080:8080 \
  -e FLASK_ENV=production \
  -e ADMIN_TOKEN=change-me \
  -e ALLOWED_ORIGINS=https://hoox.sh,https://hoox.sh/axis \
  pynescrypt-api:latest
curl -s http://127.0.0.1:8080/ | jq .
```

Compose with Redis

```
docker compose --profile redis up --build
```

Optional LSP container

```
docker compose --profile lsp run --rm lsp
```

CLI image

```
docker buildx bake cli
docker run --rm pynescrypt-cli:latest --help
docker run --rm -v "$PWD:/work" -w /work pynescrypt-cli:latest check script.pine
docker run --rm -v "$PWD:/work" -w /work pynescrypt-cli:latest lint script.pine --json
# or via compose:
make docker-cli ARGS="run script.pine --bars 50"
```

Production overlay

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml up --build -d
```



Failure modes

Symptom	Cause	Fix
Healthcheck fails	process not listening / curl missing	Use image targets from this Dockerfile; check <code>docker compose logs api</code>
Connection refused on host port	Flask bound to <code>127.0.0.1</code>	Dev target sets <code>HOST=0.0.0.0</code> ; do not override to localhost
Permission denied writing keys	Running as <code>appuser</code> outside <code>/data</code>	Set <code>API_KEY_STORE=/data/...</code> and mount <code>api_data</code>
Import errors after mount	<code>PYTHONPATH</code> / stale site-packages	<code>PYTHONPATH=/app/src:/app</code> ; rebuild after dependency changes
OOM under load	Small memory cap	Raise <code>API_MEM_LIMIT</code> / Cloud Run memory; reduce concurrency
CORS failures from PWA	Origin not allowed	Set <code>ALLOWED_ORIGINS</code>
Huge build context	Missing <code>.dockerignore</code>	Ensure <code>.dockerignore</code> is present (corpus/docs excluded)

See also

- [GCP](#)
- [Security](#)
- [Pro API lifecycle](#)
- [Local development](#)

NUITKA BUILD

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title: "Nuitka build (LSP + CLI)" description: "Compile pyne-lsp and pyne CLI with Nuitka — onefile vs standalone, CI build script, Anaconda quirks, and artifact layout."

NUITKA BUILD (LSP + CLI)

Abstract

Both the **Language Server** and the **Click CLI** can run as pure Python or as **Nuitka-compiled onefile binaries** for offline distribution. Compilation freezes the entry module and followed imports into artifacts under `dist/`.



Binary	Entry	Typical use
<code>pynescrypt-lsp</code> (PATH name; product CLI is <code>pyne-lsp</code>)	<code>langserver/__main__.py</code>	Editors / VSIX
<code>pynescrypt</code> (PATH name; product CLI is <code>pyne</code>)	<code>pynescrypt/__main__.py</code>	Shell / CI gates

Primary tools: `scripts/build/compile.py` (local) and `scripts/build/ci_build.py` (CI/Cloud Build). Both accept `--target lsp|cli|all`.

Conceptual model

```
flowchart TB
    META[generate_builtin_metadata.py] --> JSON[builtin_metadata.json]
    JSON --> ENC[Fernet encrypt → .enc + .sha256]
    JSON --> PLAIN[plaintext for dev]
    SRC[src/pynescrypt/langserver] --> NUITKA[nuitka --onefile|--standalone]
    ENC --> NUITKA
    NUITKA --> BIN[dist/.../pynescrypt-lsp]
    BIN --> VSIX[optional VSIX bundle]
```

Interface surface

Commands

```
# Prerequisites
pip install nuitka cryptography

# Full LSP onefile build (default path via Make)
make build
# ≡ python scripts/build/compile.py --target lsp --jobs=4

# CLI onefile binary
make build-cli
# ≡ python scripts/build/compile.py --target cli --jobs=4

# Fast import check only (~30s) – both targets
make build-check
# ≡ python scripts/build/compile.py --target all --check

# Explicit modes
python scripts/build/compile.py --target cli --standalone
python scripts/build/compile.py --target all --jobs 8
python scripts/build/ci_build.py --target cli --jobs=4 --skip-metadata --skip-vsix
```

Build modes (approx. wall time)

Mode	Time	Notes
<code>--check</code>	~30s	Import resolution / no full compile
<code>--standalone</code>	5–15 min	Directory distribution, faster iteration
<code>--onefile</code>	10–30 min	Self-extracting single binary
CI 8 cores	5–15 min	High parallelism



Expected layout

```
dist/
├─ lsp/
│   └─ pynescrypt-lsp          # onefile (path may vary by script)
├─ vsix/
│   └─ pynescrypt-lsp.vsix    # optional bundle
└─ pynescrypt-lsp            # ci_build final move target (script-dependent)
```

Also see `scripts/build/README.md`.

Internals

Key Nuitka flags (`compile.py`)

- `--static-libpython=no` — default; critical for many Linux/Anaconda envs
- `--python-flag=no_site,no_docstrings`
- `--include-data-dir=.../providers=pynescrypt/langserver/providers`
- `--lto=auto`, `--jobs=N`
- `--onefile` or `--standalone`
- `--follow-imports` on local compile path

Entry module

Compilation targets `src/pynescrypt/langserver` (package entry), product name `pynescrypt-lsp`.

Metadata stage

Before compile, scripts may:

1. Run `scripts/generate_builtin_metadata.py` if plaintext metadata missing
2. Fernet-encrypt to `builtin_metadata.json.enc`
3. Write truncated SHA256 sidecar
4. Write `scripts/build/.metadata.key` (gitignored) or consume `CRYPTO_KEY` env in CI

Anaconda

If static libpython is unavailable:

```
conda install libpython-static
# or rely on --static-libpython=no (already the default in compile.py)
```

Invariants & edge cases

1. **C compiler required** (gcc/clang/MSVC). Containers need `build-essential` or equivalent if compiling inside Docker.
2. **Python 3.10+**; release workflow pins **3.11** with `nuitka>=2.5.1,<2.8`. Local/Cloud Build sketches often use 3.12.
3. **Providers data dir must ship** — completion/hover collapse without metadata.



- 4. **Onfile startup cost** includes extract-to-temp; standalone is snappier for local soak tests.
- 5. **Do not commit** `scripts/build/.metadata.key`.

Worked examples

Local onfile with explicit jobs

```
python scripts/build/compile.py --jobs "$(nproc)"
find dist -name 'pynescrypt-lsp*' -type f
```

CI-shaped build

```
python scripts/build/ci_build.py --jobs=4
```

Verify without compile

```
python scripts/build/compile.py --check
```

Failure modes

Symptom	Cause	Fix
Link error on libpython	Anaconda static lib missing	Install static lib or keep <code>--static-libpython=no</code>
Binary missing builtins	Data dir not included	Confirm <code>--include-data-dir</code> providers path
Huge binary	Full stdlib pull-in	Review Nuitka plugins / unused deps
CI timeout	Low cores / onfile	Raise jobs; use larger runner; prefer standalone intermediate
Decrypt fail at runtime	Key not embedded / env unset	See Metadata crypto

See also

- [Metadata crypto](#)
- [Release](#)
- [LSP builtin metadata](#)

METADATA CRYPTO

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title: "Metadata encryption" description: "Fernet-encrypted LSP

builtin_metadata.json — key lifecycle, CI secrets, integrity hashes, and runtime decrypt."

METADATA ENCRYPTION

Abstract

LSP completion, hover, and signature help depend on a large **builtin metadata** document derived from the live evaluator surface. In open development the JSON is plaintext and git-tracked. For **compiled Nuitka binaries**, the build pipeline can ship an encrypted blob so the binary does not leave a trivially scrapable metadata file on disk.

Mechanism: **Fernet** (symmetric, `cryptography` package) + optional SHA256 integrity sidecar.



Conceptual model

```
flowchart LR
    GEN[generate_builtin_metadata.py] --> PLAIN[providers/builtin_metadata.json]
    PLAIN --> FERNET[Fernet.encrypt]
    KEY[.metadata.key or CRYPTO_KEY] --> FERNET
    FERNET --> ENC[builtin_metadata.json.enc]
    PLAIN --> SHA[builtin_metadata.json.sha256]
    ENC --> LOAD[metadata_decrypt.load_encrypted_metadata]
    KEY --> LOAD
    SHA --> LOAD
    PLAIN --> DEV[get_metadata_cached prefers plaintext]
```

Interface surface

Files

Path	Tracked?	Role
src/pynescrypt/langserver/providers/builtin_metadata.json	yes	Plaintext for dev / hatch
.../builtin_metadata.json.enc	often yes	Encrypted payload for binaries
.../builtin_metadata.json.sha256	yes	Truncated SHA256 of plaintext
scripts/build/.metadata.key	no (gitignored)	Local Fernet key material
scripts/generate_builtin_metadata.py	yes	Regenerates JSON from code

Environment variables

Variable	Context	Meaning
CRYPTO_KEY	Build scripts / CI / Cloud Build	Key bytes for encrypt stage
PYNESCRIPT_METADATA_KEY	Runtime decrypt fallback	Env-supplied key if file missing
GitHub secrets.METADATA_KEY	Actions	Stable secret → CRYPTO_KEY
Cloud Build _METADATA_KEY	cloudbuild.yaml substitution	CRYPTO_KEY=\${_METADATA_KEY}

Code entrypoints

- Encrypt stage: scripts/build/compile.py (encrypt_metadata / _resolve_fernet_key), scripts/build/ci_build.py (stage_metadata)
- Decrypt: src/pynescrypt/langserver/providers/metadata_decrypt.py
- Loader preference: plaintext if present, else encrypted (get_metadata returns {} on miss; get_metadata_cached raises)

Internals

Generation (always code-derived)

```
python scripts/generate_builtin_metadata.py
```



Do **not** hand-edit the JSON for permanent feature work — re-run the generator after adding builtins.

Encrypt (local sketch)

```
from cryptography.fernet import Fernet
key = Fernet.generate_key() # or load stable secret
fernet = Fernet(key)
encrypted = fernet.encrypt(plaintext_bytes)
```

Build scripts chmod the key file to `0o600` when writing `.metadata.key`.

Decrypt integrity check

After Fernet decrypt, if a `.sha256` sidecar exists, the loader compares `sha256(plaintext)[:16]` and raises on mismatch.

Dev vs binary

Mode	Behavior
Editable install / repo checkout	Prefer plaintext JSON
Nuitka onefile	Encrypted data dir + key resolution via MEIPASS / env
Missing both	<code>FileNotFoundError</code> directing to compile script

Invariants & edge cases

1. **Stable key ⇒ reproducible `.enc`**. Random key every CI run yields a different ciphertext even if plaintext is identical — not a functional bug, but pollutes diffs and caches.
2. **Plaintext remains in git** for open-source transparency of the *language surface*. Encryption protects the *bundled binary layout*, not the secret of the API catalog.
3. **Truncated hash is integrity, not secrecy**. It detects bitrot / wrong plaintext, not attackers with the key.
4. **Never commit `.metadata.key`**. Rotate if leaked; regenerate `.enc` with the new key for binary builds.
5. **After adding builtins**: regenerate metadata, re-encrypt with the **same** CI key, commit updated JSON (and `.enc` if tracked).

Worked examples

Regenerate + local encrypt

```
python scripts/generate_builtin_metadata.py
pip install cryptography
python scripts/build/compile.py --check # stages metadata paths as configured
```



CI snippet

```
- name: Build LSP binary
  run: python scripts/build/ci_build.py --jobs 4
  env:
    CRYPTO_KEY: ${ secrets.METADATA_KEY }
```

Runtime with env key

```
pynescrypt-lsp
```

Failure modes

Symptom	Cause	Fix
No metadata decryption key found	Missing file + env	Supply PYNESCRYPT_METADATA_KEY or rebuild with key
Metadata integrity check failed	Stale hash or wrong ciphertext	Re-encrypt from current plaintext; commit both
Completion empty in binary only	Data dir not included in Nuitka	Fix --include-data-dir
Divergent .enc every CI	Unset CRYPTO_KEY	Configure METADATA_KEY secret

See also

- Nuitka build
- LSP builtin metadata
- Release
- Security

GCP

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title: "GCP & Cloud Run" description: "Cloud Build pipeline, Cloud Run Pro API sizing, cost model, and substitution secrets for PYNE deployments."

GCP & CLOUD RUN

Abstract

Production-shaped hosting for the Pro API targets **Google Cloud Run**, fed by **Cloud Build** (`cloudbuild.yaml`). The pipeline builds a container, pushes multi-tags to Container Registry, deploys a managed service, and can optionally compile the LSP binary in the same build graph.

This page codifies the as-checked-in config and the cost envelope from `docs/gcp_cost_estimate.md` — treat dollar figures as planning estimates, not invoices.



Conceptual model

```
flowchart LR
    GIT[push main] --> CB[Cloud Build]
    CB --> DOCKER["docker build --target api"]
    DOCKER --> GCR[gcr.io project/repo/service]
    GCR --> RUN[Cloud Run deploy]
    CB --> LSP[optional Nuitka ci_build]
    RUN --> SQL[(Cloud SQL optional)]
    RUN --> REDIS[(Memorystore optional)]
    RUN --> GCS[Cloud Storage charts]
```

Interface surface

Substitutions (`ccloudbuild.yaml`)

Substitution	Default	Meaning
<code>_SERVICE_NAME</code>	<code>pynescrypt-pro-api</code>	Cloud Run service
<code>_REGION</code>	<code>us-central1</code>	Deploy region
<code>_REPOSITORY</code>	<code>pynescrypt</code>	Image path segment
<code>_METADATA_KEY</code>	<code>(secret/subst)</code>	Fernet key → <code>CRYPTO_KEY</code> for LSP stage
<code>_PYNESCRIPT_VERSION</code>	<code>"0.6.7"</code>	OCI label. Matches <code>src/pynescrypt/__about__.py</code> .

Build steps

1. **build** — `docker build` with BuildKit; tags `$COMMIT_SHA + latest` ; cache-from `latest`
2. **push** — `docker push --all-tags`
3. **deploy** — `gcloud run deploy` with managed platform flags
4. **build-lsp** — `python:3.12-slim` image runs `scripts/build/ci_build.py --jobs=4` with `CRYPTO_KEY=${_METADATA_KEY}`



Cloud Run flags (as configured)

Flag	Value	Rationale
<code>--allow-unauthenticated</code>	on	Public HTTP API (auth via API keys at app layer)
<code>--min-instances</code>	0	Scale to zero
<code>--max-instances</code>	10	Cap spend / blast radius
<code>--memory</code>	512Mi	Evaluator headroom
<code>--cpu</code>	1	Single vCPU per instance
<code>--concurrency</code>	10	Parallel requests per instance
<code>--timeout</code>	60s	Platform kill. Entrypoint gunicorn default is 120s (<code>GUNICORN_TIMEOUT</code>); set the env to 60 on Cloud Run or long evals die at the platform first.
<code>--set-env-vars</code>	<code>FLASK_ENV=production</code>	App mode

Machine / budget knobs

```
options:
  machineType: E2_HIGHCPU_8
  logging: CLOUD_LOGGING_ONLY
  timeout: 1200s
```

Internals

Path	Role
<code>cloudbuild.yaml</code>	Pipeline definition
<code>Dockerfile</code> <code>target</code> <code>api</code>	Production image recipe used by Cloud Build
<code>backend/app.py</code>	Health <code>/</code> , CORS, body size limit
<code>docs/gcp_cost_estimate.md</code>	Scenario costs (April 2026 draft)

Cost envelope (planning)

Scale	Rough monthly	Notes
Free / hobby	~\$0	Within Cloud Run free tier if tiny
MVP (~10 pro users)	~\$50	Cloud Run + micro SQL
Growth (~50)	~\$120	Add Redis caching
Scale (~200)	~\$350	Dominated by evaluator CPU-sec

Dominant cost driver: **script execution CPU time** (~2–5 CPU-sec per typical backtest over 1k bars). Memory ~200–400MB per concurrent Python worker.



Free-tier leverage (GCP)

- Cloud Run: 2M requests, 400K CPU-sec, 200K GB-sec / month
- Cloud Build: 120 build-minutes / day
- Artifact Registry / GCS small free slices

Invariants & edge cases

1. **App-layer auth still required** when Cloud Run allows unauthenticated invoke — see [Security](#).
2. **Scale-to-zero cold starts** add latency; min-instances=1 trades money for snappiness.
3. **LSP build step does not deploy the binary** to Cloud Run; it is a release-adjacent compile in the same YAML.
4. **512Mi may be tight** for large OHLCV payloads — watch OOM kills; raise memory before raising concurrency.
5. **Cost estimate doc** also lists Railway / Render / Fly / self-host alternatives if GCP is overkill.

Worked examples

Manual deploy sketch (operator)

```
gcloud builds submit --config cloudbuild.yaml \  
  --substitutions=_METADATA_KEY="$METADATA_KEY"
```

Smoke health after deploy

```
curl -s "https://SERVICE-URL/"  
# expect JSON status healthy, service pynescrypt-pro-api
```

Failure modes

Symptom	Cause	Fix
Deploy timeout	Nuitka step + docker on 20 min budget	Split LSP compile to release pipeline
429 / queue	max-instances or concurrency	Raise caps carefully; add queue
High bill	No cache; chatty clients	Redis TTL; Cloud Tasks for backtests
Auth bypass illusion	Unauthenticated Cloud Run	Enforce API keys in Flask
Image pull errors	Wrong project / registry perms	IAM on GCR + Cloud Run service agent

See also

- [Docker](#)
- [Observability](#)
- [Security](#)
- [Pro API](#)

OBSERVABILITY

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title: "Observability" description: "Health endpoints, gunicorn process model, Cloud Logging, coverage artifacts, and operational signals for PYNE."

OBSERVABILITY

Abstract

PYNE's production surface today optimizes for **simple, hostable signals** rather than a full OpenTelemetry mesh: HTTP health checks, process managers (gunicorn), cloud platform logs, CI artifacts (coverage, Playwright reports), and application-level API key usage counters. Treat this page as the map of *what exists in-repo*, not a claim of enterprise APM parity.



Conceptual model

```
flowchart TB
    CLIENT[Client] --> LB[Cloud Run / local]
    LB --> GUNI[gunicorn workers]
    GUNI --> APP[Flask backend.app]
    APP --> HEALTH["GET /"]
    APP --> RUN["POST /run ..."]
    APP --> KEYS[APIKeyStore counters]
    GUNI --> LOGS[stdout / Cloud Logging]
    CI[GitHub Actions] --> ART[coverage xml / VSIX / docker smoke]
```

Interface surface

Health

backend/app.py exposes:

```
GET /
→ { "status": "healthy", "service": "pynescrypt-pro-api", ... }
```

Used by:

- Dockerfile target api HEALTHCHECK (curl -fsS http://127.0.0.1:8080/)
- docker-compose.yml api healthcheck
- Load balancers / Cloud Run startup-ish probes (platform-dependent)

Process model (production)

```
gunicorn --bind :8080 --workers 2 --threads 4 --timeout 120 backend.app:app
# docker/entrypoint-api.sh: GUNICORN_TIMEOUT default 120
# docker-compose.prod.yml: GUNICORN_TIMEOUT default 60
```

Knob	Effect on signals
workers	Process isolation; multiplies memory
threads	Concurrent requests within worker
timeout	Entrypoint default 120s . Cloud Run <code>--timeout=60s</code> is tighter unless you raise the platform flag or lower <code>GUNICORN_TIMEOUT</code> .

Logs

Environment	Where logs go
Local <code>make run</code>	Process stdout/stderr
Docker	Container logs (<code>docker logs</code> , <code>compose</code>)
Cloud Build	<code>logging: CLOUD_LOGGING_ONLY</code>
Cloud Run	Cloud Logging (request + container)



Application modules use standard library / Flask logging patterns; there is no mandatory structured-log schema enforced repo-wide. `` if a future branch introduces OpenTelemetry exporters.

CI / quality signals

Signal	Source
Ruff / mypy	CI lint job
Pytest results	runtime + LSP + backend steps in CI
Codecov (optional)	Python 3.13 matrix cell
VSIX	CI vscode-ext-test artifact
Docker smoke	CI api health + admin 403; CLI --help / info / check

Usage metering (app-layer)

backend/middleware/auth.py tracks per-key:

- calls_used / calls_limit / tier
- last_used timestamps

These are **business metrics** for rate limits, not Prometheus time series — export them if you need dashboards.

Internals

Path	Role
backend/app.py	Health route, CORS, MAX_CONTENT_LENGTH
backend/middleware/auth.py	Key store, rate limit counters
backend/services/backtest.py	Backtest metrics objects for API responses
Dockerfile (api target)	HEALTHCHECK + gunicorn entrypoint
cloudbuild.yaml	Cloud Logging-only build logs
logs/ (repo)	Local combined/error log files if generated by tools — not the production contract

Invariants & edge cases

1. **Health ≠ readiness of evaluator warm caches.** A 200 on / does not mean the first /run will be fast (cold import / JIT / numba).
2. **Timeouts double-bind.** Cloud Run 60s vs gunicorn default 120s — the **platform** limit wins on Cloud Run unless you set GUNICORN_TIMEOUT=60 .
3. **Scale-to-zero hides idle metrics.** No traffic ⇒ no samples; use synthetic uptime checks if SLOs matter.
4. **Coverage artifacts expire** (7–14 days retention on Actions uploads).
5. **Do not scrape secrets from logs.** API keys and admin tokens must never be logged at info level.



Worked examples

Local health loop

```
make run
curl -s http://127.0.0.1:5002/ | python -m json.tool
```

Docker health

```
docker inspect --format='{{json .State.Health}}' CONTAINER
```

Tail Cloud Run (gcloud)

```
gcloud run services logs read pynescrypt-pro-api --region us-central1 --limit 50
```

Failure modes

Symptom	Cause	Fix
Health green, <code>/run</code> 500	Exception in evaluator path	Inspect request logs; reproduce with payload
Worker kills	Soft timeout	Optimize script / raise timeout consistently
Missing CI artifact	Path wrong / tests passed	<code>if-no-files-found: ignore</code> hides absence
Rate limit false positives	Shared key / limit too low	Adjust tier limits; multi-key
Silent deploy failure	Build logging-only + unread	Check Cloud Build history UI

See also

- [GCP](#)
- [Docker](#)
- [Security](#)
- [API lifecycle](#)

SECURITY

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title: "Security" description: "API keys, CORS, body limits, secrets hygiene, Fernet metadata keys, and threat-aware defaults for PYNE ops."

SECURITY

Abstract

PYNE security is layered: **transport/hosting** (Cloud Run, containers), **application** (API keys, admin tokens, CORS, body size caps), **build secrets** (Fernet metadata keys, VSCE PAT), and **supply chain** (CI dependency installs, extension packaging). This page documents controls that exist in code and workflows — not a full threat model certification.



Conceptual model

```
flowchart TB
    NET[Internet] --> CR[Cloud Run allow-unauthenticated]
    CR --> CORS[flask-cors allowlist]
    CORS --> SIZE[MAX_CONTENT_LENGTH 5MB]
    SIZE --> AUTH[require_api_key]
    AUTH --> STORE[APIKeyStore hash + tiers]
    AUTH --> RUN[Evaluator /run]
    ADMIN[X-Admin-Token] --> KEYMGMT[key minting routes]
    BUILD[CI secrets] --> META[METADATA_KEY]
    BUILD --> VSCE[VSCE_PAT]
```

Interface surface

HTTP application controls (`backend/app.py`)

Control	Default / mechanism
Body size	<code>MAX_CONTENT_LENGTH = 5 * 1024 * 1024</code> (5 MiB) — DoS/OOM mitigation
CORS	<code>ALLOWED_ORIGINS</code> env (comma-separated); default includes product hosts + localhost regex
Methods	GET, POST only via flask-cors config
Headers	<code>Content-Type</code> , <code>Authorization</code> , <code>X-Admin-Token</code>
Credentials	<code>supports_credentials=False</code>

API keys (`backend/middleware/auth.py`)

- Keys stored as **hashes**, not raw secrets (store path via `API_KEY_STORE`)
- Tiers: `free`, `hobby` (5k), `pro` (50k), `team` (200k), `enterprise` (∞)
- Rate limit by monthly call counters (`calls_used` / `calls_limit`)
- Decorators: `require_api_key`, `require_admin_token`

Build / release secrets

Secret	Purpose
<code>METADATA_KEY</code> / <code>_METADATA_KEY</code>	Stable Fernet key for LSP metadata encrypt
<code>VSCE_PAT</code>	VS Code Marketplace / vsce package
<code>GITHUB_TOKEN</code>	Release asset upload
<code>PYNESCRYPT_METADATA_KEY</code>	Runtime decrypt env fallback

Container hardening (`Dockerfile` target `api`)

- Multi-stage build (no compiler toolchain in final image)
- Non-root `appuser` (uid 1000)
- Production env flags + gunicorn entrypoint
- Healthcheck without privileged ops



- Intended mutable path `/data` (key store); prod compose uses SQLite there for multi-worker

Admin minting

- `POST /auth/create_key` requires `ADMIN_TOKEN` env **and** matching `X-Admin-Token` header (constant-time compare)
- Unset `ADMIN_TOKEN` → 403 (fail closed, not open access)

Key store backends (`STORE_BACKEND`)

Value	Use
<code>json</code> (default)	Single-process / tests; file may hold raw keys
<code>sqlite</code>	Multi-worker single host (shared volume); hash-only
<code>redis</code>	Multi-replica; needs <code>REDIS_URL</code> ; hash-only

Internals

Path	Role
<code>backend/app.py</code>	CORS, body limit, blueprint registration
<code>backend/middleware/auth.py</code>	Key store selection, tiers, decorators, admin gate
<code>backend/middleware/key_store_redis.py</code> / <code>key_store_sqlite.py</code>	Hash-only shared stores
<code>backend/middleware/schemas.py</code>	Request validation schemas
<code>src/pynescrypt/langserver/providers/metadata_decrypt.py</code>	Encrypted metadata load
<code>.github/workflows/*.yaml</code>	Secret consumption (<code>METADATA_KEY</code> , <code>PYPI_API_TOKEN</code> , <code>VSCE_PAT</code> , <code>OVSX_PAT</code>)
AXIS PWA security tests	Sister repo hoox-sh/axis — not in this tree

Audit notes encoded in source

Comments in `app.py` reference a **2026-07-05 audit** (findings S1 body size, S7/S8 CORS). Prefer reading the code over assuming defaults never change.

Invariants & edge cases

1. **Cloud Run `--allow-unauthenticated` does not mean open data.** It means IAM does not gate invoke; the Flask app must.
2. **Free tier with `calls_limit=0` is special-cased** as unlimited remaining in some helpers — verify tier semantics before production billing.
3. **Localhost CORS regex** permits any port on localhost/127.0.0.1 — fine for desk, wrong for prod `ALLOWED_ORIGINS` .
4. **Fernet protects binary metadata packaging**, not user script confidentiality.
5. **Admin token routes are high privilege** — rotate and store separately from user API keys.
6. **Never log raw API keys.** Hash lookups only.



Worked examples

Harden CORS for a single origin

```
python -m backend.app
```

Point key store at a volume

```
# ensure process user can write /data
```

Rotate metadata key (ops procedure sketch)

- 1. Generate new Fernet key; store as secret `METADATA_KEY`
- 2. Rebuild LSP with `CRYPTO_KEY` set
- 3. Ship new binary; retire old key after client upgrade window
- 4. Do not leave dual-key decrypt unless you implement it

Failure modes

Symptom	Cause	Fix
Browser CORS error	Origin not listed	Set <code>ALLOWED_ORIGINS</code>
413 Payload Too Large	Body > 5MB	Reduce OHLCV batch; raise limit knowingly
401/403 on <code>/run</code>	Missing/invalid key	Check <code>Authorization</code> header scheme
Rate limited	Tier exhausted	Upgrade tier / wait reset
Secret in fork PR logs	Misconfigured workflow echo	Never <code>echo</code> secrets; use masked env
Root-owned files in volume	Old image ran as root	Align UID with <code>appuser</code>

See also

- [Auth and keys \(API\)](#)
- [Metadata crypto](#)
- [Docker](#)
- [GCP](#)